

Permanent and induced magnetism

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 12 starred (★) questions first.

	QUESTION	ANSWER	✓
1	What does 'attract' mean?	A force pulls the two objects towards each other.	☐
2	What does 'repel' mean?	A force pushes the two objects away from each other.	☐
3	Name the two ends of a magnet.	The north pole and the south pole	☐
4	State what a north pole does to another north pole.	It repels it.	☐
5	State what a north pole does to a south pole.	It attracts it.	☐
6 ★	Name three magnetic chemical elements.	Iron, nickel and cobalt	☐
7 ★	Name an alloy that is magnetic.	Steel	☐
8 ★	What is a permanent magnet?	A magnet that produces its own magnetic field.	☐
9 ★	What is an induced magnet?	A material that becomes a magnet when placed in a magnetic field.	☐
10 ★	Is the force between an induced magnet and a permanent magnet attractive or repulsive?	Always attractive	☐
11	State what surrounds a permanent magnet.	A magnetic field	☐
12	Describe how the strength of a magnetic field changes with distance from the magnet.	It gets weaker as the distance increases.	☐

	QUESTION	ANSWER	✓
13 ★	State where the field of a permanent magnet is strongest.	At the poles.	☐
14 ★	State how field lines show where a magnetic field is strongest.	The lines are closest together.	☐
15 ★	State the direction in which magnetic field lines point.	From the north pole to the south pole.	☐
16	Describe what a compass contains.	A small bar magnet that is free to rotate.	☐
17	Give two ways to show the shape of the magnetic field around a magnet.	Plotting compasses or iron filings	☐
18 ★	Describe how to find the direction of the magnetic field around a magnet.	Use a plotting compass: its needle points along the field, from north to south.	☐
19	Name the pole at the end of a compass needle that points north.	A north pole	☐
20 ★	State the evidence that the Earth's core is magnetic.	A compass needle lines up north–south when no other magnet is near, so the Earth has its own magnetic field.	☐
21 ★	What is a magnetic field?	The region around a magnet where a force acts on another magnet or magnetic material.	☐
22	Is the force between two magnets a contact or a non-contact force?	Non-contact	☐
23	State what happens to an induced magnet when it is removed from the field.	It quickly loses its magnetism.	☐
24 ★	State how the direction of a magnetic field is defined.	The direction of the force on a north pole placed at that point.	☐